

MODULE 09

Elbow MRI: Ligaments, Nerves, OCD & Tendon Injuries

UCL Grading · Capitellar OCD · Lateral/Medial Epicondylosis · PLRI · Ulnar Nerve · Distal Biceps & Triceps

1.0 AMA PRA Cat 1

Elbow

CME Article Review

LEARNING OBJECTIVES

1. Grade ulnar collateral ligament (UCL) injuries on coronal MRI using established imaging criteria and the T-sign.
2. Distinguish stable from unstable osteochondral lesions (OCD) of the capitellum and explain the management implications of each stage.
3. Identify lateral epicondylosis (common extensor origin / ECRB) and medial epicondylosis (common flexor-pronator origin) on MRI, including grading by tendon signal and fiber disruption.
4. Recognize posterolateral rotatory instability (PLRI) and posteromedial rotatory instability (PMRI), and their relationship to lateral ulnar collateral ligament (LUCL) and coronoid disruption.
5. Interpret ulnar nerve cross-sectional area (CSA) at the cubital tunnel and identify imaging findings of cubital tunnel syndrome, including dynamic subluxation.
6. Diagnose distal biceps and triceps tendon tears on axial and sagittal MRI, including localization for surgical planning.

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1. Ulnar Collateral Ligament (UCL) Injury Grading

The ulnar collateral ligament (UCL) complex — sometimes still called the medial collateral ligament (MCL) of the elbow in older literature — has three components: the anterior bundle, the posterior bundle, and the transverse (Cooper) ligament. The **anterior bundle** is the primary restraint to valgus stress throughout the arc of flexion and is the structure of overwhelming clinical interest, particularly in overhead throwing athletes (baseball pitchers, javelin throwers) who generate enormous repetitive valgus torque during the late cocking and acceleration phases of the throwing motion. The anterior bundle originates from the inferior surface of the medial epicondyle and inserts on the sublime tubercle of the proximal ulna; it is best assessed on coronal proton-density or T2 fat-saturated sequences, with the elbow extended.

Repetitive microtrauma produces a spectrum from ligamentous attenuation and partial undersurface tearing to frank complete rupture. Because partial tears are far more common than complete tears in throwers, and because management decisions hinge on distinguishing partial from complete injury, the radiologist must specifically interrogate the deep (undersurface) fibers, where the classic **T-sign** is sought.

UCL Grading — Coronal PD / T2 Fat-Saturated MRI

Grade	Pathology	MRI Findings	Management
I — Sprain	Microtears; fibers grossly intact	Periligamentous and intraligamentous edema; ligament thickened but continuous; no discontinuity	Conservative; structured rehabilitation and activity modification
II — Partial tear	Partial-thickness fiber disruption, usually involving the deep (undersurface) fibers	Positive T-sign — contrast or fluid extends deep to the distal ligament in a T-shaped configuration on coronal images; partial discontinuity; intraligamentous T2 hyperintensity	Conservative vs. platelet-rich plasma (PRP) injection vs. surgical repair, depending on athlete demands and level of competition
III — Complete tear	Full-thickness rupture	Complete fiber discontinuity with a fluid-filled gap; most commonly proximal at the medial epicondyle origin, less commonly distal at the sublime tubercle, least commonly midsubstance; medial joint space widening on valgus stress views	UCL reconstruction ("Tommy John surgery") is typically required for competitive overhead athletes to return to play

Reporting tip: always state the location of the tear (proximal at the epicondyle, distal at the sublime tubercle, or midsubstance), explicitly identify which bundle is involved (the anterior bundle is by far the most clinically significant), and comment on any associated medial epicondyle bone marrow edema, avulsion fragment, or flexor-pronator mass injury, since combined injuries change the surgical approach.

Adjunct UCL MRI Measurements

Measurement	Normal Value	Clinical Significance
UCL thickness (coronal)	≤ 3–4 mm	Thickening > 5 mm suggests chronic tear with scar/fibrosis formation
Medial (ulnohumeral) joint space (coronal, valgus stress)	2–4 mm	Widening > 4–5 mm relative to the contralateral side suggests medial instability
Common flexor-pronator origin signal	Homogeneous low T1/T2 signal	Increased T2 signal indicates concurrent medial epicondylosis ("golfer's elbow"), commonly co-existing with UCL injury

PEARL

The T-sign is the single most useful sign for an undersurface partial UCL tear and is best seen on coronal MR arthrography, where contrast tracks deep to the distal ligament fibers in a T-shaped pattern at the sublime tubercle. On conventional (non-arthrographic) MRI, look for an analogous focus of fluid signal in the same location.

REFERENCES

- Timmerman LA, Andrews JR. Undersurface tear of the ulnar collateral ligament in baseball players: a newly recognized lesion. *Am J Sports Med* 1994.
- Schwartz ML et al. Ulnar collateral ligament injury in the throwing athlete: evaluation with saline-enhanced MR arthrography. *Radiology* 1995.

2. Osteochondritis Dissecans (OCD) of the Capitellum

Osteochondral lesions of the capitellum occur in the lateral compartment of the elbow and arise from repetitive compressive and shear loading between the radial head and capitellum, which is characteristic of overhead throwing and weight-bearing sports such as gymnastics. The typical patient is an **adolescent athlete, roughly 10–15 years old** — distinguishing this from Panner disease, which affects a younger population (5–10 years) and represents a self-limited osteonecrosis of the entire capitellar ossific nucleus rather than a focal osteochondral lesion. MRI is the imaging modality of choice for staging because management is fundamentally driven by **fragment stability**, which radiographs cannot reliably assess.

Capitellar OCD Staging — MRI (Coronal / Sagittal T1, T2/STIR)

Stage	Cartilage / Subchondral Bone	Stability	Management
I	Cartilage intact; subchondral marrow edema/signal change only; no fragmentation	Stable	Conservative (rest, activity modification)
II	Cartilage breach with an underlying fracture line, but the fragment remains in situ without fluid undermining	Stable (early)	Conservative trial; close follow-up for progression
III	Fragment partially detached; thin rim of high-T2-signal fluid undermines part of the fragment-bone interface	Unstable (incipient)	Often requires surgical fixation or debridement depending on size
IV	Loose intra-articular osteochondral body; complete detachment with an empty crater at the donor site	Unstable	Surgical — fragment fixation, removal, or osteochondral autograft/allograft transfer

The single most important determinant of stage and management is whether **fluid undermines the fragment** at the bone–cartilage interface on fluid-sensitive sequences — this finding indicates instability regardless of the degree of surface cartilage disruption, and its presence should be explicitly stated in the report since it directly changes the treatment pathway from conservative rest to surgical referral.

PITFALL

Do not confuse capitellar OCD with Panner disease. Panner disease affects younger children (roughly 5–10 years), involves the entire capitellar ossification center diffusely, and is self-limited with a good prognosis on conservative management alone. Capitellar OCD affects older adolescents, is focal, and carries a real risk of progression to loose body formation and premature osteoarthritis if unstable lesions go untreated.

REFERENCES

- Takahara M et al. Classification, treatment, and outcome of osteochondritis dissecans of the humeral capitellum. *J Bone Joint Surg Am* 2007.
- Kijowski R, De Smet AA. Radiography of the elbow for evaluation of patients with osteochondritis dissecans of the capitellum. *Skeletal Radiol* 2005.

3. Lateral and Medial Epicondylitis

Despite the long-entrenched terms "lateral epicondylitis" (tennis elbow) and "medial epicondylitis" (golfer's elbow), histopathology in both conditions shows angiofibroblastic tendinosis rather than true inflammation, so the preferred terminology is **epicondylitis**. Lateral epicondylitis involves the common extensor origin, with the extensor carpi radialis brevis (ECRB) tendon as the primary and most consistently affected structure; medial epicondylitis involves the common flexor-pronator origin. Both conditions are graded using an identical framework based on intratendinous signal change and the degree of fiber disruption.

Epicondylitis Grading (Applies to Both Lateral [ECRB] and Medial [Common Flexor] Origins)

Grade	Pathology	MRI Findings
1 — Tendinosis	Mucoid degeneration without fiber disruption	Increased intratendinous signal on fluid-sensitive sequences; tendon contour and fiber continuity preserved
2 — Partial tear	Partial-thickness fiber disruption, < 50% of tendon cross-section	± peritendinous fluid; focal fiber discontinuity involving less than half the tendon width
3 — High-grade / near-complete tear	Partial-thickness involving ≥ 50% of cross-section	Larger fiber gap; associated peritendinous edema and fluid; tendon substantially thinned
4 — Complete tear	Full-thickness tendon disruption	Discontinuous tendon with retraction and a fluid-filled gap at the epicondyle

Two associated findings deserve specific mention in every report. First, on the lateral side, assess the lateral collateral ligament complex (specifically the LUCL) deep to the common extensor origin, since combined ECRB and LUCL injury raises concern for posterolateral rotatory instability (see Section 4). Second, on the medial side, medial epicondylitis coexists with ulnar neuropathy in roughly a quarter of cases, so the cubital tunnel and ulnar nerve should be specifically assessed whenever medial epicondylitis is identified.

CLINICAL TIP

Lateral epicondylitis and a LUCL tear can occur together after the same traumatic event (such as a fall on an outstretched, varus-stressed arm) or develop together from chronic overuse. When lateral epicondylitis is identified, always specifically comment on LUCL integrity rather than limiting the report to the tendon alone.

REFERENCES

- Potter HG et al. Lateral epicondylitis: correlation of MR imaging, surgical, and histopathologic findings. *Radiology* 1995.
- Kijowski R, De Smet AA. MRI findings of medial epicondylitis. *AJR Am J Roentgenol* 2005.

4. Posterolateral and Posteromedial Rotatory Instability

Rotatory instability of the elbow is a spectrum of subluxation injuries distinct from simple valgus or varus laxity. **Posterolateral rotatory instability (PLRI)** is the most common pattern and results from disruption of the lateral ulnar collateral ligament (LUCL), the critical sling-like stabilizer that prevents the radial head and ulna from rotating posteriorly and laterally away from the trochlea, classically following an elbow dislocation or chronic varus/posterolateral stress. Clinically, patients have a positive pivot-shift (lateral pivot-shift) test. On coronal and axial MRI, look for discontinuity of the LUCL at its origin on the lateral epicondyle, frequently with associated lateral epicondylitis of the common extensor origin immediately superficial to it — both structures should be assessed whenever one is injured.

Posteromedial rotatory instability (PMRI) is a less common but increasingly recognized pattern involving a fracture of the anteromedial facet of the coronoid combined with disruption of the LUCL, producing instability in a varus, posteromedial rotatory direction. The O'Driscoll classification of anteromedial coronoid facet fractures is used to grade severity and guide surgical planning.

O'Driscoll Classification — Anteromedial Coronoid Facet Fracture (PMRI)

Subtype	Fracture Extent	Clinical Implication
1	Anteromedial rim only	May be stable; smaller fragment
2	Anteromedial rim plus the tip of the coronoid	Larger fragment; increasing instability risk
3	Anteromedial rim extending into the base of the coronoid process	Most severe; high likelihood of frank instability requiring fixation, particularly when combined with LUCL injury

CLINICAL TIP

For PLRI, the key MRI finding is LUCL discontinuity at the lateral epicondyle, often paired with lateral epicondylitis — assess both structures together. For PMRI, identify the anteromedial coronoid facet fracture pattern using the O'Driscoll system and specifically evaluate LUCL integrity, since the combination of these two injuries defines the unstable pattern requiring surgical fixation.

REFERENCES

- O'Driscoll SW, Jupiter JB, Cohen MS et al. Difficult elbow fractures: pearls and pitfalls. *Instr Course Lect* 2003.
- Sanchez-Sotelo J et al. Posteromedial rotatory instability of the elbow. *J Bone Joint Surg Am* 2005.

5. Ulnar Nerve and Cubital Tunnel Syndrome

The ulnar nerve passes through the cubital tunnel posterior to the medial epicondyle, bounded by the Osborne ligament (the fibrous arcade connecting the two heads of the flexor carpi ulnaris). This is the most common site of ulnar nerve compression in the upper extremity. MRI evaluation should assess nerve caliber, signal, position, and the downstream flexor carpi ulnaris muscle for evidence of denervation.

Ulnar Nerve MRI Assessment at the Cubital Tunnel

Parameter	Normal	Abnormal Threshold	Significance
Cross-sectional area (axial, at the tunnel)	$\leq 9\text{--}10\text{ mm}^2$	$> 10\text{--}12\text{ mm}^2 = \text{enlarged}$	Enlargement supports compressive neuropathy / cubital tunnel syndrome
Signal intensity (axial T2 fat-sat)	Homogeneous; isointense to other peripheral nerves	Increased intrinsic T2 signal	Suggests intraneural edema or early demyelination
Flexor carpi ulnaris muscle signal	Normal muscle signal	T2 hyperintensity (acute) or fatty atrophy (chronic)	Denervation edema acutely, or fat replacement chronically, both confirm a neurogenic process
Position relative to the medial epicondyle	Stable, within the tunnel	Anterior subluxation or frank dislocation over the epicondyle with flexion	"Snapping" ulnar nerve; may itself cause a friction neuritis independent of fixed compression

A thickened, enlarged ulnar nerve with increased T2 signal at the retroepicondylar groove or just distal to it within the cubital tunnel proper is the classic finding of entrapment. Dynamic subluxation is best assessed with the elbow imaged in both extension and flexion when clinically suspected, since a nerve that is normally positioned in extension may sublux anteriorly over the medial epicondyle only with flexion.

REFERENCES

- Bordalo-Rodrigues M, Rosenberg ZS. MR imaging of entrapment neuropathies at the elbow. *Magn Reson Imaging Clin N Am* 2004.
- Britz GW et al. Ulnar nerve entrapment at the elbow: correlation of magnetic resonance imaging, electrodiagnostic, and intraoperative findings. *Neurosurgery* 1996.

6. Distal Biceps and Triceps Tendon Tears

Distal biceps tendon tears occur almost exclusively in middle-aged men during an eccentric loading event (the classic mechanism is a forceful contraction against resistance, such as catching a heavy falling object). The tendon inserts on the radial tuberosity, and the **axial plane at the level of the radial tuberosity** is the primary plane for evaluating tear continuity, supplemented by sagittal images to assess the degree of proximal retraction for surgical planning.

Distal Biceps Tear Classification — Axial MRI at the Radial Tuberosity

Type	Finding	Key Imaging Clue
Tendinopathy	Thickened tendon with intratendinous T2 signal; no discrete fiber gap	Fusiform T2 hyperintensity at the radial tuberosity insertion; the posterior (bursal-surface) fibers are most often affected first
Partial tear	Partial fiber discontinuity with fluid signal within the tendon substance	Most commonly involves the posterior (bursal-surface) fibers; assess remaining fiber continuity carefully since a partial tear can mimic a complete tear if not traced on multiple planes
Complete tear	Full-thickness discontinuity; tendon not identifiable at the radial tuberosity	"Empty tuberosity" sign on axial images; proximal tendon stump retraction seen on sagittal; assess the lacertus fibrosus, since an intact lacertus limits retraction and changes the surgical approach

PEARL

For distal biceps tears, use the axial plane at the radial tuberosity as the primary evaluation plane. A complete tear produces the "empty tuberosity" sign on axial images. Always report the degree of proximal retraction on sagittal images for surgical planning, and specifically comment on whether the lacertus fibrosus (bicipital aponeurosis) remains intact, since an intact lacertus can tether the tendon and limit retraction despite a complete tear at the tuberosity.

Distal triceps tendon tears are far less common than distal biceps tears but follow an analogous pattern of evaluation: the triceps inserts on the olecranon, and tears typically occur at or near this osseous insertion following a forced flexion of an extended, contracting elbow (a fall onto an outstretched hand is a classic mechanism) or, less commonly, a direct blow. Sagittal images best demonstrate the tendon gap and degree of retraction, while axial images confirm whether any residual fibers (often the deep articular fibers) remain intact, which can represent a partial-tear pattern that does not always require operative repair.

A bony avulsion fragment from the olecranon ("flake sign" on radiographs) frequently accompanies triceps avulsions and should be specifically searched for, since its presence confirms an osseous avulsion mechanism and can guide fixation technique. Bicipitoradial bursitis (fluid distending the bursa between the distal biceps tendon and the radial tuberosity) may also be seen with chronic distal biceps tendinopathy and should be distinguished from a tear itself.

REFERENCES

- Giuffre BM, Moss MJ. Optimal positioning for MRI of the distal biceps brachii tendon: flexed abducted supinated view. *AJR Am J Roentgenol* 2004.
- Potter HG et al. Magnetic resonance imaging of the elbow: an update. *Clin Sports Med* 2004.